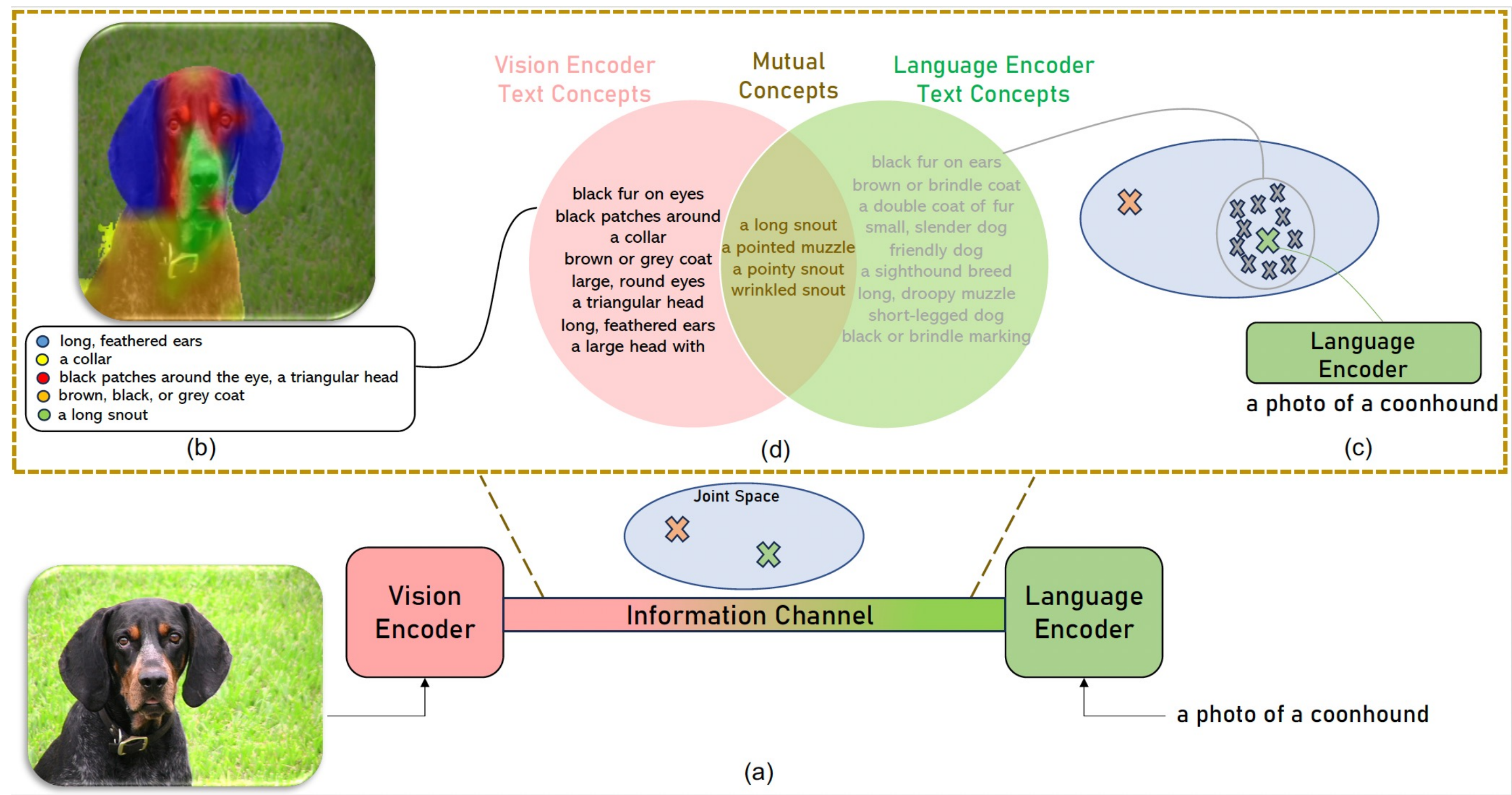


Interpreting and Analyzing CLIP's Zero-Shot Image Classification via Mutual Knowledge

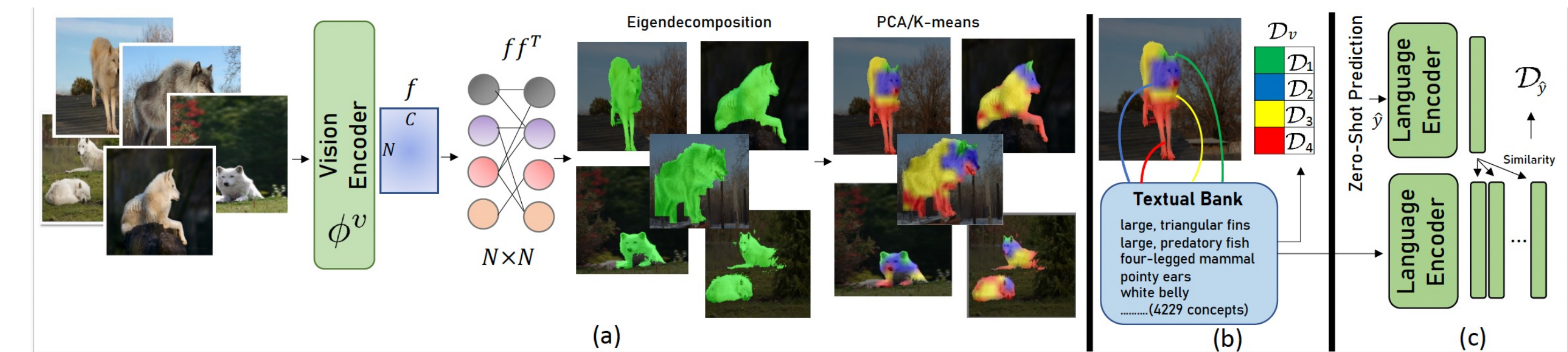
Fawaz Sammani, Nikos Deligiannis



What concepts do the vision and language encoders of CLIP learn in common, causing image-text points to be closer or further apart in the joint space?

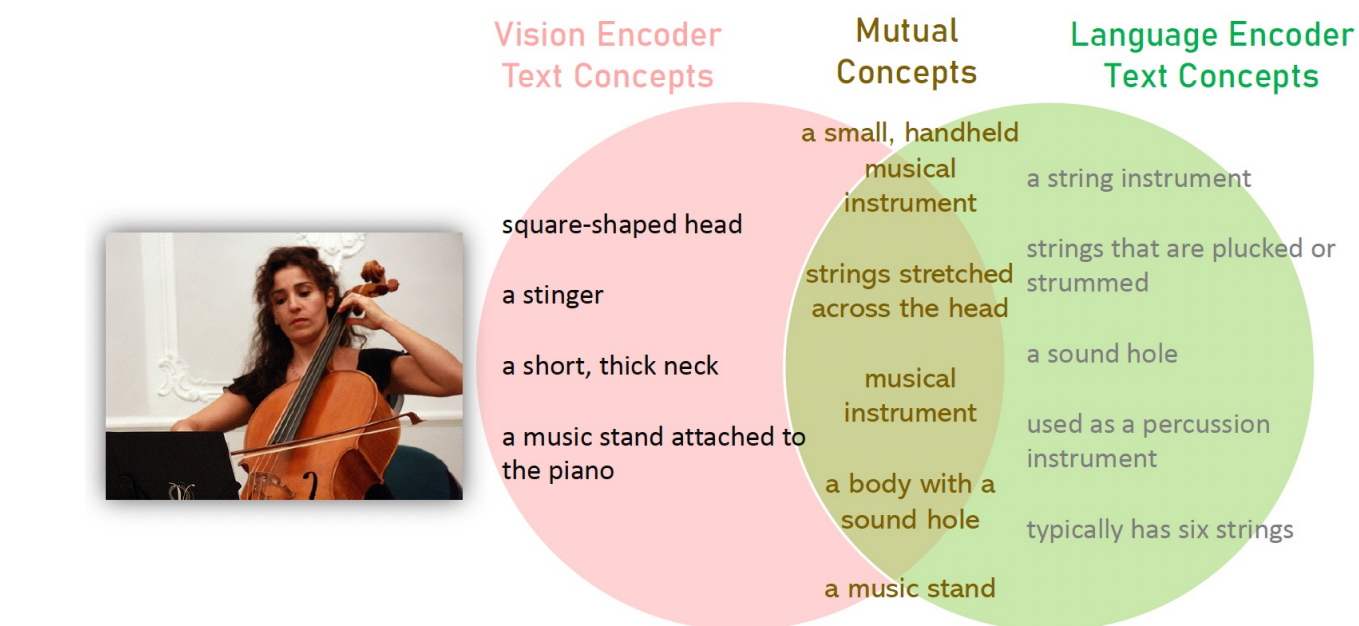


Method



$$I(\mathcal{D}_v; \mathcal{D}_{\hat{u}}) = H(\mathcal{D}_v) + H(\mathcal{D}_{\hat{u}}) - H(\mathcal{D}_v, \mathcal{D}_{\hat{u}})$$

More Examples



Why Textual Concepts

- Human-Friendly Interpretations
- Allows for a shared interpretation medium between vision and language encoders
- Discrete Units, Easy and interpretable Mutual Information calculation

Multimodal Concepts

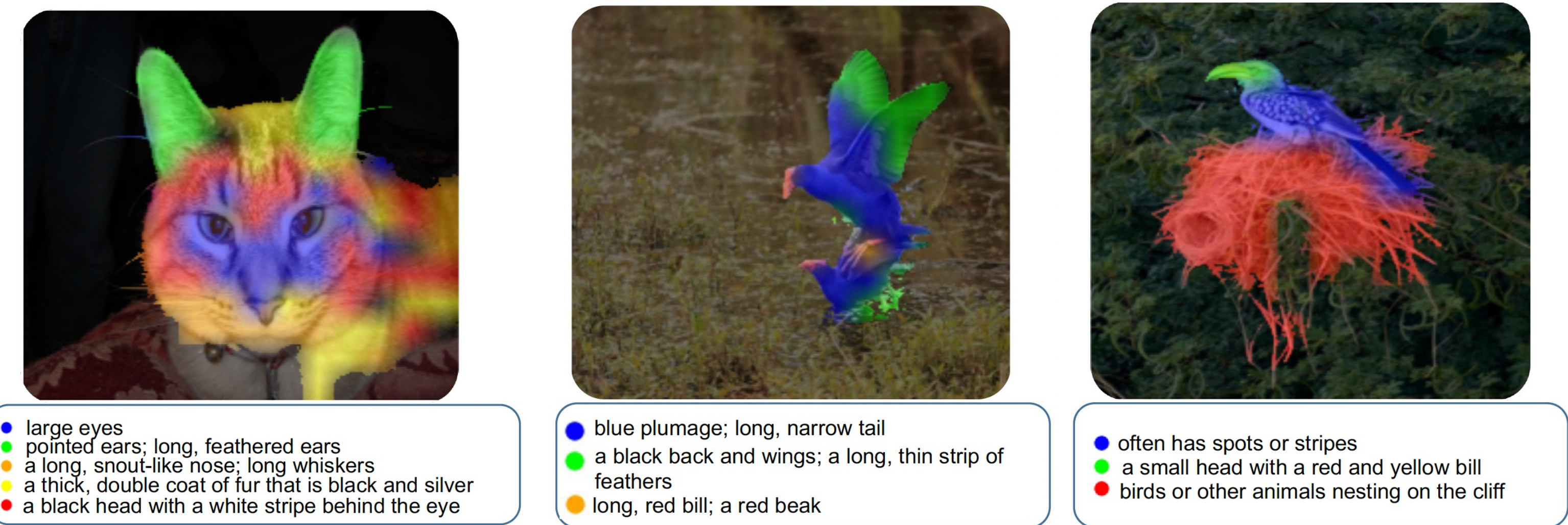


Table 2: Effectiveness and Relevancy of our multimodal concepts in boosting zero-shot accuracy of both ResNet and ViT CLIP models on the ImageNet validation set compared to baselines [36, 43].

ResNets	Base	Ours	Δ	ViTs	Base	Ours	Δ
RN50	59.54	61.85	+2.31	ViT-B/16	67.93	70.28	+2.35
RN50x4	64.36	67.93	+3.57	ViT-B/32	63.28	65.58	+2.30
RN50x16	68.47	72.22	+3.75	ViT-L/14	74.69	76.74	+2.05
RN101	60.68	64.14	+3.46	ViT-L/14@336px	75.49	77.64	+2.15

CLIP Models Analysis

Table 3: MI and AUC scores for different model families using PCA and K-means evaluated on the full ImageNet validation split, along with the pretraining data and Top-1 accuracy.

Model Family	Model	Data Size	Top-1 (%)	MI		AUC	
				PCA	K-means	PCA	K-means
ViTs	ViT-B/32	400M	61.66	7.40	7.26	3.61	3.39
	ViT-B/16	400M	67.70	7.50	7.44	3.62	3.53
	ViT-B/32-dcp	1B	68.88	7.79	7.65	3.93	3.70
	ViT-B/16-dcp	1B	73.37	7.68	7.58	3.99	3.81
	ViT-L/14	400M	74.77	7.94	7.89	4.47	4.37
	ViT-L/14 $\uparrow$	400M	76.23	7.96	7.93	4.51	4.44
	ViT-B/16-dfn	2B	76.24	8.19	8.11	4.62	4.46
ResNets	RN-50	400M	58.42	7.14	7.20	3.23	3.32
	RN-101	400M	60.90	7.43	7.53	3.49	3.60
	RN-50 $\times$ 4	400M	65.28	7.53	7.58	3.84	3.90
	RN-50 $\times$ 16	400M	70.04	7.51	7.63	3.85	4.03
ConvNeXTs	CNeXt-B1	400M	65.36	6.47	6.66	2.54	2.80
	CNeXt-B2	13B	71.22	7.16	7.56	3.19	3.74